

HW 5 con.

4c. Proof Method 2:

$$A \oplus B = (A \cap B^c) \cup (A^c \cap B)$$

$$r_1 = A \cap B^c \cap C^c$$

$$r_2 = A^c \cap B \cap C^c$$

$$r_3 = A^c \cap B^c \cap C$$

$$r_4 = A \cap B \cap C^c$$

$$r_5 = A \cap B^c \cap C$$

$$r_6 = A^c \cap B \cap C$$

$$r_7 = A \cap B \cap C$$

$$r_8 = A^c \cap B^c \cap C^c$$

$$(A \oplus B) \oplus C = ((A \cap B^c) \cup (A^c \cap B)) \oplus C$$

$$= ((A \cap B^c) \cap C^c) \cup ((A \cap B^c)^c \cap C) \cup$$

$$((A^c \cap B) \cap C^c) \cup ((A^c \cap B)^c \cap C)$$

$$(A \oplus B) \oplus C = r_1 \cup r_2 \cup r_3 \cup r_7 \quad \checkmark$$

$$A \oplus (B \oplus C) = A \oplus ((B \cap C^c) \cup (B^c \cap C))$$

$$A \oplus (B \oplus C) = r_1 \cup r_2 \cup r_3 \cup r_7 \quad \checkmark$$

Equivalent: • Both sides expand to the same Union of the same disjoint intersections.

$$(A \oplus B) \oplus C = A \oplus (B \oplus C)$$